

Geography Ch-2 — The Earth's Crust (Rocks)

Physical Geography | Petrology / Rock Systems | GS-I

FOUNDATION: G.C. Leong · STANDARD: Majid Husain (Indian & World Geography)

Register-style revision notes · grounded in source books

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — The Three Rock Families (G.C. Leong)

■ NEWS TRIGGER

The basaltic **Deccan Traps** — a classic extrusive-rock plateau — recur in prelims (soil, aquifer, mineral) and in news on black-soil cotton belts and basalt weathering.

■ INTRO & ORIGIN

The crust is built of three rock families defined by how they form: **igneous** (from cooled magma), **sedimentary** (from accumulated sediment) and **metamorphic** (pre-existing rock altered by heat & pressure).

*Igneous is the 'primary' rock; its debris builds sedimentary rock; heat & pressure convert either into metamorphic rock — the essence of the **rock cycle**.*

■ KEY DATA

Family	Sub-type	Examples / features
Igneous — Plutonic (intrusive)	Slow cooling at depth → LARGE crystals	Granite, diorite, gabbro
Igneous — Volcanic (extrusive)	Rapid cooling at surface → SMALL crystals	Basalt (Deccan Plateau, Giant's Causeway)
Sedimentary — Mechanical	Cemented rock fragments	Sandstone (quartz grains)
Sedimentary — Organic	Remains of organisms	Limestone, chalk (corals/shells)
Metamorphic	Heat + pressure alter parent rock	Marble, slate, quartzite, gneiss

■ STATIC THEORY

- **Plutonic (intrusive) rocks** cool slowly deep in the crust → large, easily recognised crystals (granite); exposed later by denudation.
- **Volcanic (extrusive) rocks** pour out as lava and solidify fast → tiny crystals; basalt forms lava sheets/plateaux (Deccan, Columbia-Snake, Iceland).
- **Sedimentary rocks are stratified** (layered) — usually deposited under water; layers vary from inches to many feet.
- Sediment is transported by streams, glaciers, wind or even animals before deposition and cementation.
- **Metamorphic transformations** under great heat/pressure: clay→slate, limestone→marble, sandstone→quartzite, granite→gneiss, shale→schist, coal→graphite.

■ MUST-KNOW FACTS

1. Igneous = cooled magma; plutonic (large crystals) vs volcanic (small crystals).
2. Basalt = extrusive → Deccan Traps; columnar basalt = Giant's Causeway (Antrim).
3. Sedimentary rocks = STRATIFIED (layered), mostly water-laid.

■ UPSC TRAPS

- ✗ Basalt cools slowly to form large crystals.
- ✓ Basalt is EXTRUSIVE → rapid cooling → small crystals; granite (plutonic) has large crystals.
- ✗ All sedimentary rocks come from river sand.

4. Limestone/chalk = organically formed (corals, shells); sandstone = mechanically formed.

5. Metamorphic key pairs: limestone→marble, sandstone→quartzite, coal→graphite.

6. Granite→gneiss; shale→schist; clay→slate.

✓ They can be mechanical (sandstone), organic (limestone/chalk) or chemical; sediment may be river-, wind-, glacier- or organism-derived.

✗ Marble is an igneous rock.

✓ Marble is METAMORPHIC — from limestone under heat & pressure.

■ MAINS ANGLE

Rock type controls soil, aquifer and mineral wealth — e.g. Deccan basalt → black cotton soil; a base for linking geology to agriculture and resource geography.

■ STUDY LINK

Geography → Physical → Rocks / Petrology (Leong Ch-2)

HIGH

STANDARD — Rock Classification, Characteristics & the Rock Cycle (Majid Husain)

GS-I ·

Geomorphology

→ Petrology

(Source: Majid

Husain, Indian &

World

Geography)

■ NEWS TRIGGER

Foliation and perviousness of metamorphic rock (marble, slate, gneiss) govern dam-site stability and groundwater potential — recurring in engineering-geology and aquifer news.

■ INTRO & ORIGIN

Majid Husain classifies **igneous rocks by texture and composition** — basalt, gabbro, andesite, diorite, rhyolite, granite — and by mode of occurrence into **intrusive (plutonic)** and **extrusive (volcanic)**.

Igneous rock is the most important original source of the inorganic mineral matter that later forms sedimentary rock — so the three families are genetically linked in the rock cycle.

■ KEY DATA

Metamorphic rock	Parent rock / grade	Diagnostic character
Slate	Shale (low-grade)	Excellent foliation = 'slaty cleavage'
Gneiss	Granite (high-grade regional)	Coarse, BANDED quartz + feldspar
Marble	Limestone	Crystalline; impervious
Quartzite	Sandstone	Hard; source rock easily identified
Schist	Shale / mica-rich	Splits along bedding planes (mica-schist)

■ STATIC THEORY

- **Sedimentary characteristics:** derived from pre-existing rock (igneous the chief source) + organic matter; always layered/stratified.
- Sub-aerially exposed riverine sedimentary rocks develop **polygonal desiccation cracks** (mud cracks) on drying.
- **Glacial sedimentary deposits** = moraines; wind-laid fine silt = loess (e.g. northern China).
- **Metamorphic rocks** form from pre-existing rock by change in temperature, pressure and chemical action of fluids — altering texture AND mineral composition.

- Perviousness varies: marble & slate are **impervious**; gneiss can be **pervious** — important for groundwater and foundations.
- Gneiss shows granular quartz-feldspar banding; slate shows slaty cleavage — both aid field identification.

■ MUST-KNOW FACTS

1. Igneous rocks classified by TEXTURE + COMPOSITION (basalt, gabbro, andesite, diorite, rhyolite, granite).
2. Igneous rock = chief original source of sedimentary mineral matter.
3. Slate = low-grade metamorphism of shale (slaty cleavage).
4. Gneiss = high-grade regional metamorphism; banded quartz-feldspar.
5. Marble & slate impervious; gneiss can be pervious.
6. Loess = wind-deposited silt; moraine = glacial sediment.

■ UPSC TRAPS

- ✗ Foliation/cleavage and stratification are the same.
- ✓ Foliation (slate, schist) is a metamorphic alignment; stratification is sedimentary layering — different origins.
- ✗ All metamorphic rocks are impervious.
- ✓ Marble & slate are impervious, but gneiss can be pervious — perviousness varies by rock.
- ✗ Source rock of every metamorphic rock is obvious.
- ✓ For quartzite/marble it is clear; for schist and gneiss the parent is often hard to determine.

■ MAINS ANGLE

Discuss how rock genesis and properties (perviousness, foliation, crystallinity) determine engineering suitability, aquifers and mineralisation — connecting petrology to resource planning.

■ STUDY LINK

Geography → Geomorphology → Rock systems & the Rock Cycle (Majid Husain)